

What is Third Normal Form (3NF)?

Definition

A relation is in Third Normal Form (3NF) if:

- It is already in Second Normal Form (2NF).
- It contains no Transitive Dependencies.

Formal Definition

For every Functional Dependency:

$$X \rightarrow A$$

At least one must be true:

- X is a Super Key, OR
- A is a Prime Attribute (part of some Candidate Key).

Converting to 3NF

Original

EmployeeID	EmployeeName	DepartmentID	DepartmentName
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FDs

EmployeeID $\rightarrow$ EmployeeName
EmployeeID $\rightarrow$ DepartmentID
DepartmentID $\rightarrow$ DepartmentName

Step 1

Identify the transitive dependency.

DepartmentID → **DepartmentName**

Step 2

Create a separate Department table.

Employee

EmployeeID	EmployeeName	DepartmentID
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Department

DepartmentID	DepartmentName
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Now each fact is stored only once.

Dependency Preservation

Definition

A decomposition is Dependency Preserving if all original Functional Dependencies can still be enforced without joining tables.

If dependencies are not preserved:

- More joins are needed to validate data.
- Updates become more expensive.

3NF allows one exception:

A dependency is acceptable if the dependent attribute is a Prime Attribute.

Sometimes this still leaves redundancy.

BCNF removes this exception.

BCNF Rule

A relation is in BCNF if:

Every determinant is a Candidate Key.

Why Isn't 3NF Always Enough?

Consider

Teacher	Course	Room
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FDs

Teacher → **Room**

Course → **Teacher**

Suppose each teacher has one room.

Each course has one teacher.

Candidate Keys

Course

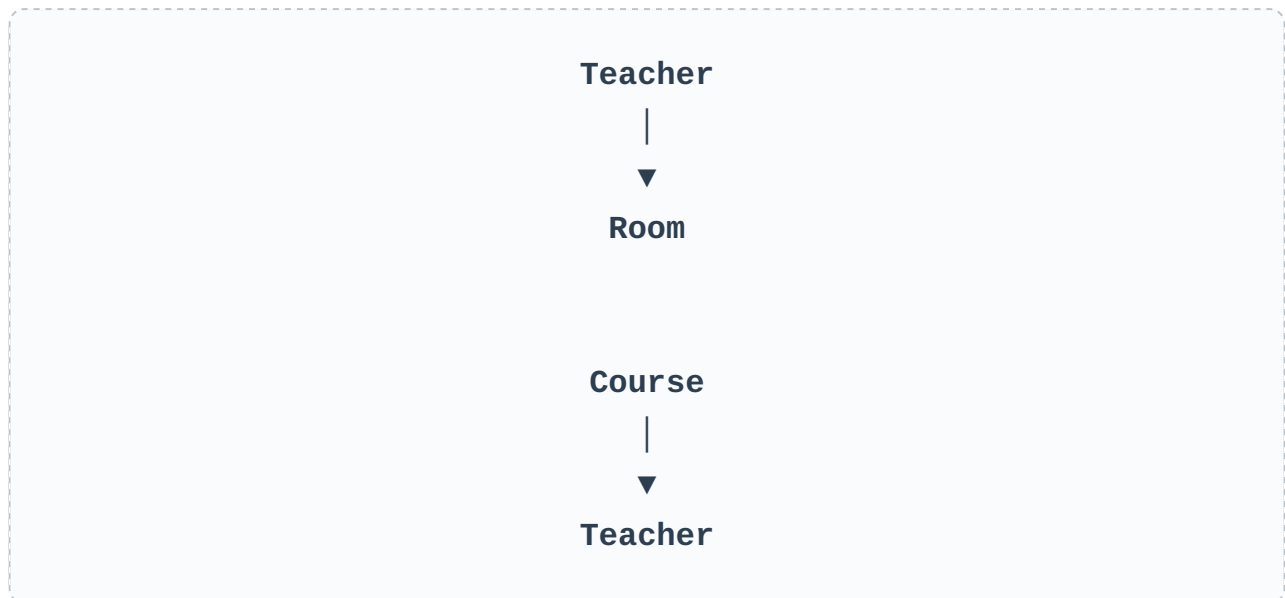
Since

Teacher → **Room**

Teacher is not a Candidate Key.

The table violates BCNF.

Yet it can satisfy 3NF depending on the key structure and prime attributes.



Teacher determines Room but is not a Candidate Key.

Violation.

BCNF Algorithm

1. Find Functional Dependencies.
2. Find Candidate Keys.
3. Find any FD where the determinant is not a Candidate Key.
4. Decompose the relation.
5. Repeat until every relation satisfies BCNF.

BCNF sometimes lose Dependency Preservation?

Answer

BCNF focuses on removing all violations where a determinant is not a Candidate Key. During decomposition, some original Functional Dependencies may become split across multiple tables. Checking or enforcing those dependencies may require joining tables, so the decomposition is not dependency preserving.

***3NF** is often preferred when preserving dependencies is important.*

***BCNF** is preferred when eliminating redundancy is more important.*